DSA06 - Data Handling and Visualization

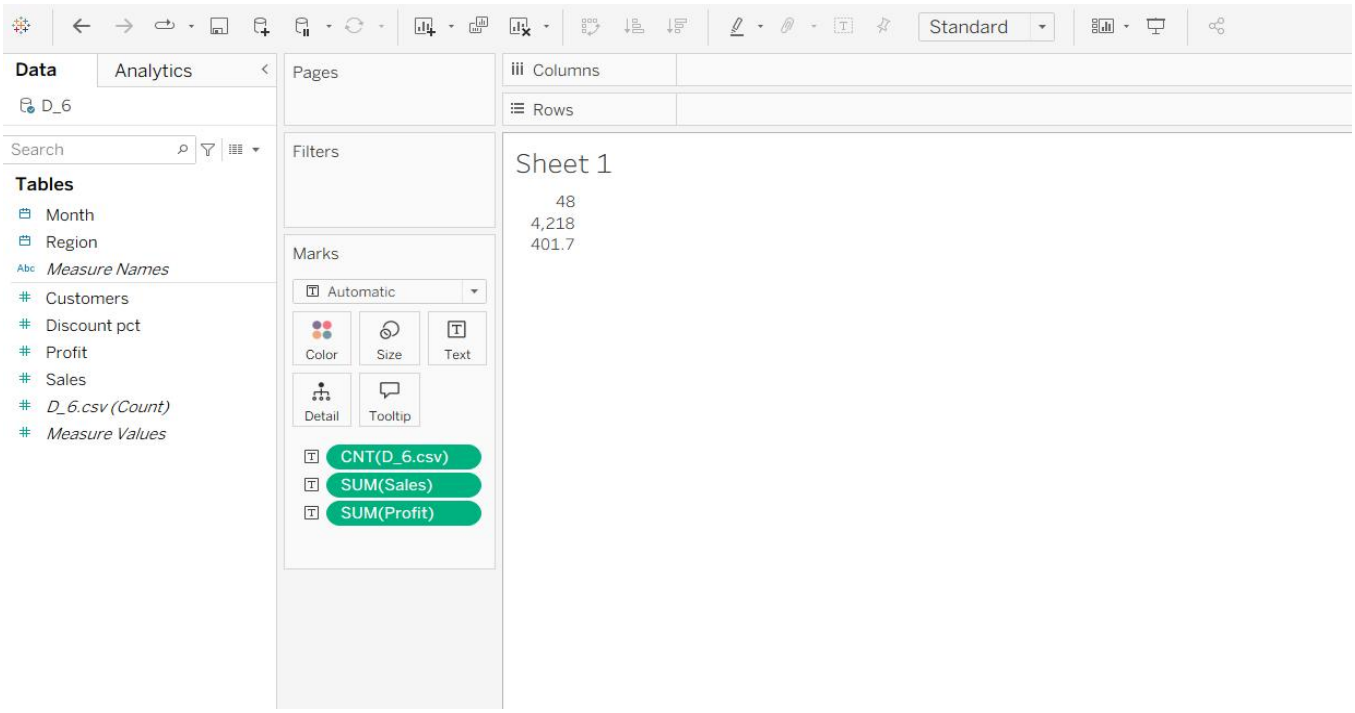
Tableau Introduction Activity

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Course Code / Name	DSA0602 / Data Handling and Visualization

Scenario: The file holds 12 months of 2025 for four regions (Telangana, Kerala, Maharashtra, Andhra Pradesh). Fields: Month, Region, Sales (Rs lakh), Profit (Rs lakh), Customers, Discount_pct. All figures below are calculated from this CSV file.

T1. Connect and check

Connect to the CSV (Text file), set Month as a Date. Report rows, dimensions and measures, total Sales and total Profit.



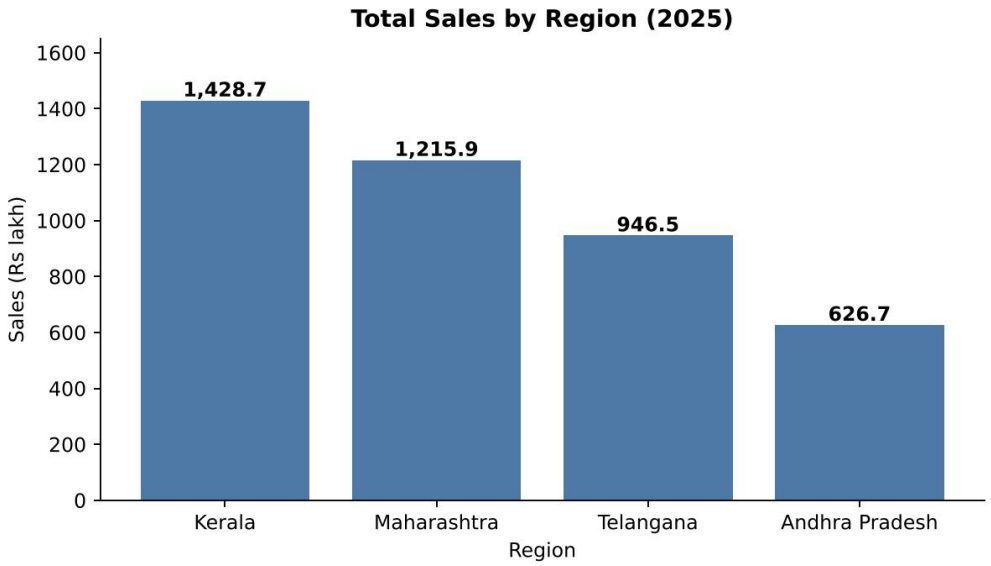
Output 1: Data source and Data pane after setting Month to Date

Item	Value
Number of rows	48 (12 months x 4 regions)
Dimensions	Month (Date), Region (String)
Measures	Sales, Profit, Customers, Discount_pct
Total Sales	4,217.8 lakh
Total Profit	401.7 lakh

Ans: The file has 48 rows and 6 columns. Month and Region are dimensions (Month was changed to Date type); Sales, Profit, Customers and Discount_pct are numeric measures. Total Sales is Rs 4,217.8 lakh and total Profit is Rs 401.7 lakh (overall margin 9.52%).

T2. Bar chart

Bar chart of Sales by Region, sorted highest to lowest, with labels. Name top and lowest region and the top region's share of total sales.



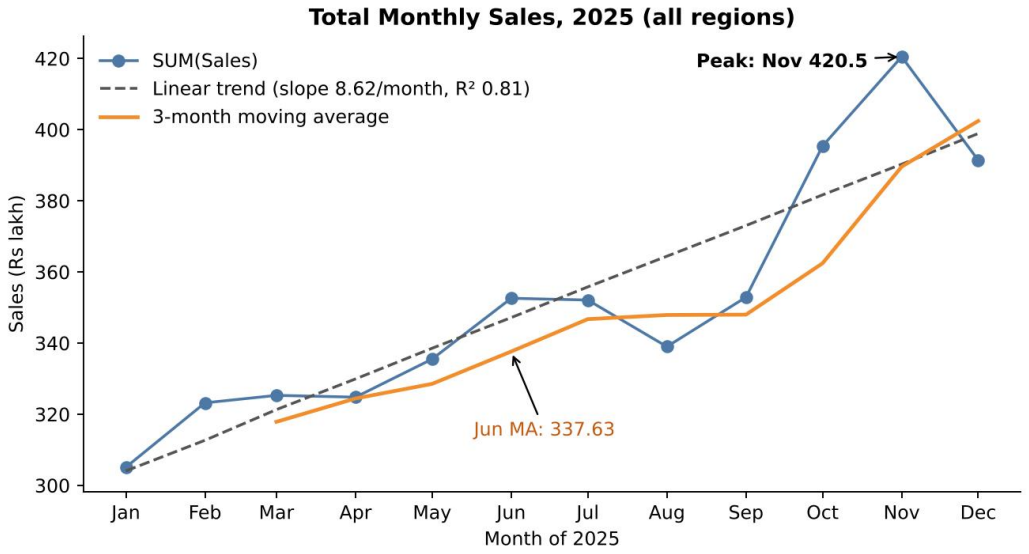
Output 2: Total Sales by Region (sorted descending, labels on bars)

Region	Sales (lakh)	Share of total
Kerala	1,428.7	33.87%
Maharashtra	1,215.9	28.83%
Telangana	946.5	22.44%
Andhra Pradesh	626.7	14.86%
Total	4,217.8	100%

Ans: The top region is Kerala (1,428.7 lakh) and the lowest is Andhra Pradesh (626.7 lakh). Kerala's share of total sales is 1,428.7 / 4,217.8 = 33.87%, about one third of all sales.

T3. Line chart and trend

Line chart of total Sales by Month with a linear trend line and a 3-month moving average. Report the peak month, the moving average at June and the trend slope per month.



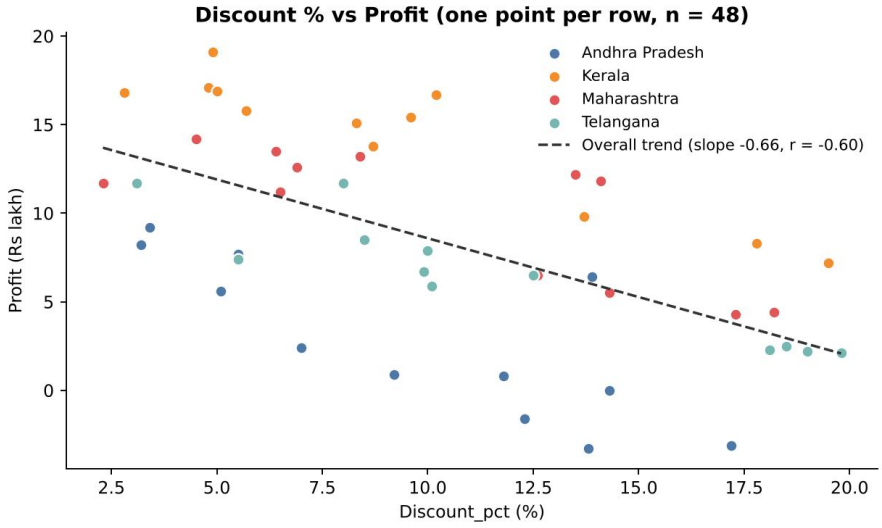
Output 3: Monthly total Sales with linear trend line and 3-month moving average

Measure	Value
Peak month	November 2025 - 420.5 lakh
3-month moving average at June	337.63 lakh (Apr 324.8, May 335.5, Jun 352.6)
Trend slope (Describe Trend Model)	+8.62 lakh per month (R-squared = 0.81)

Ans: Sales peak in November 2025 at 420.5 lakh. The moving average at June is (324.8 + 335.5 + 352.6) / 3 = 337.63 lakh. The linear trend rises by about 8.62 lakh every month, so total sales grew steadily through the year, with a sharp jump in October and November.

T4. Scatterplot

Scatterplot of Discount_pct (columns) against Profit (rows), one point per row, coloured by Region, with one overall trend line. Report slope, r and what it means.



Output 4: Discount_pct vs Profit, coloured by Region, overall trend line

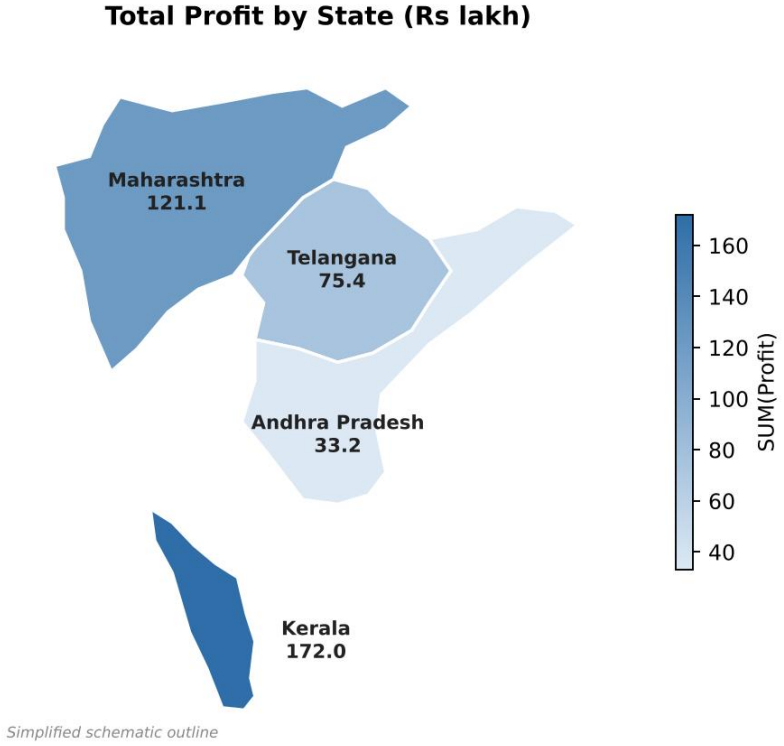
Measure	Value
Slope	-0.66 lakh of profit per 1 percentage point of discount

R-squared	0.356
r (sign of slope)	-0.60 (moderate negative)

Ans: The slope is -0.66, R-squared is 0.356, so r = -0.60. For the business, higher discounts go with lower profit: each extra percentage point of discount is linked to about 0.66 lakh less profit, so heavy discounting is eroding margins.

T5. Filled map

Filled map of Profit by Region. Name the region with the lowest total Profit and compute its profit margin.



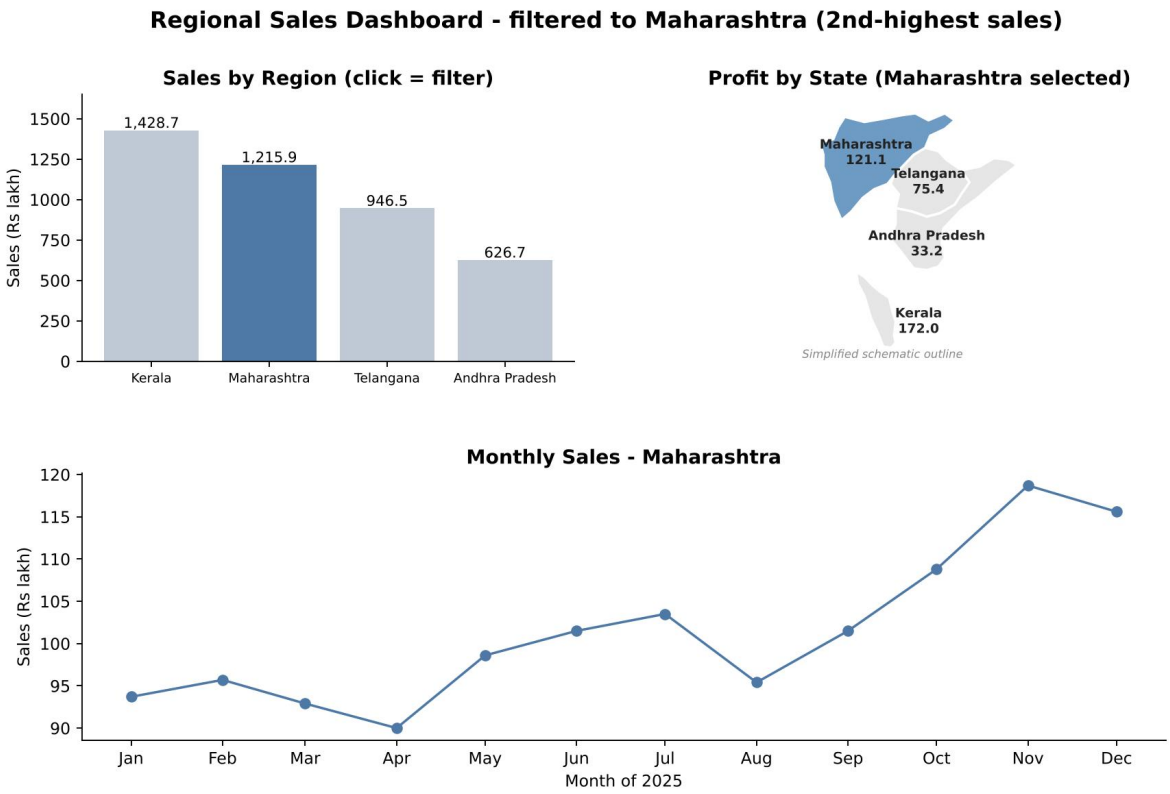
Output 5: Filled map of total Profit by state (darker = higher profit)

Region	Profit (lakh)	Sales (lakh)	Margin
Kerala	172.0	1,428.7	12.04%
Maharashtra	121.1	1,215.9	9.96%
Telangana	75.4	946.5	7.97%
Andhra Pradesh	33.2	626.7	5.30%

Ans: Andhra Pradesh has the lowest total Profit at 33.2 lakh. Its profit margin is 33.2 / 626.7 x 100 = 5.30%, the lowest of the four regions.

T6. Dashboard with a filter

Bar chart, line chart and map on one dashboard with the bar chart as a filter. Click the region with the SECOND-highest sales and report its Sales, Profit and margin. Publish and paste the link.



Output 6: Dashboard after clicking Maharashtra in the bar chart (Use as Filter)

Measure	Maharashtra
Total Sales	1,215.9 lakh
Total Profit	121.1 lakh
Profit margin	121.1 / 1,215.9 = 9.96%

Ans: The second-highest sales region is Maharashtra (Kerala is first). With the bar chart set to Use as Filter and Maharashtra clicked, the dashboard shows Sales = 1,215.9 lakh, Profit = 121.1 lakh and a profit margin of 9.96%.

T7. Compare with R

Two real differences between making the T3 line chart in Tableau and in R (ggplot2).

Ans: 1) Aggregation: in Tableau, dragging Month and SUM(Sales) onto the view adds up the four regions automatically. In R the data must first be grouped and summed in code (group_by(Month) then summarise(Sales = sum(Sales))) before ggplot2 can draw one line.

2) Trend and moving average: in Tableau these are added with clicks (Analytics pane > Trend Line, and Quick Table Calculation > Moving Average), and Describe Trend Model reports the slope and R-squared directly. In R they must be coded: geom_smooth(method = "lm") for the trend, zoo::rollmean(Sales, 3, fill = NA, align = "right") for the moving average, and summary(lm()) to read the slope. The R script is repeatable on new data, while the Tableau steps are done by hand each time.

Example R code:

```
library(ggplot2); library(dplyr); library(zoo)
m <- read.csv("D_6.csv") %>% mutate(Month = as.Date(Month)) %>%
     group_by(Month) %>% summarise(Sales = sum(Sales)) %>%
     mutate(MA3 = rollmean(Sales, 3, fill = NA, align = "right"))
ggplot(m, aes(Month, Sales)) + geom_line() + geom_point() +
  geom_smooth(method = "lm", se = FALSE, linetype = "dashed") +
  geom_line(aes(y = MA3), colour = "orange") +
  labs(title = "Total Monthly Sales, 2025", x = "Month", y = "Sales (Rs lakh)")
```

